

Department of Aerospace Engineering

JGI Global Campus, Jakkasandra Post, Kanakapura Taluk, Ramanagara District, Pin Code: 562 112
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End Year Project

Report on

“HIGH ALTITUDE LONG ENDURANCE (HALE) UNMANNED AERIAL VEHICLES (UAVs): SUSTAINED FLIGHT THROUGH RENEWABLE ENERGY”

Submitted in partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY IN AEROSPACE ENGINEERING

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CERTIFICATE

This is to certify that the project work titled "HIGH ALTITUDE LONG ENDURANCE (HALE) UNMANNED AERIAL VEHICLES (UAVs): SUSTAINED FLIGHT THROUGH RENEWABLE ENERGY" has been carried out by Parth Patil (22BTRAS031), a bonafide student of the Bachelor of Technology programme at the Faculty of Engineering & Technology, JAIN (Deemed-to-be University), Bengaluru, in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Aerospace Engineering during the academic year 2025–2026.

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DECLARATION

I, Parth Patil (22BTRAS031), a student of the 8th Semester, B.Tech in Aerospace Engineering at the Faculty of Engineering & Technology, JAIN (Deemed-to-be University), hereby declare that the End Year Project Report titled "HIGH ALTITUDE LONG ENDURANCE (HALE) UNMANNED AERIAL VEHICLES (UAVs): SUSTAINED FLIGHT THROUGH RENEWABLE ENERGY" has been carried out by me and submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Aerospace Engineering during the academic year 2025–2026.

I further declare that the work presented in this report is original and has not been submitted previously, either in part or in full, for the award of any degree or diploma at this or any other university, to the best of my knowledge and belief.

Signature

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Date: 17th February 2026

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Signature of Students

Abstract

High Altitude Long Endurance (HALE) Unmanned Aerial Vehicles (UAVs) operate at altitudes above 18 km with mission durations from several days to weeks. This project focuses on the conceptual design, performance analysis, and feasibility assessment of a solar-powered HALE UAV. Key design challenges include low Reynolds number aerodynamics, lightweight composite structures, and hybrid renewable energy integration. Simulations and theoretical modelling are employed to validate performance assumptions and analyse energy storage limitations.

This report broadens the literature review to include recent developments in solar-powered UAV design, propulsion optimisation, photovoltaic cell integration onto composite wings, battery-free UAV architectures, aeroelastic stability modelling, and hybrid turbofan–solar power concepts. The numerical evaluation indicates that the conceptual HALE model produces realistic power demands (~ 2 kW electrical), while high-altitude solar flux combined with triple-junction gallium arsenide (GaAs) photovoltaics provides ~ 9 kW usable power. Night-time storage requirements (~ 23.9 kWh) make pure lithium-ion battery solutions infeasible (~ 95 kg), whereas hydrogen fuel-cell storage meets the requirement at < 5 kg system mass.

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1. Introduction

HALE UAVs bridge the gap between conventional aircraft and satellites, remaining aloft for days or weeks at altitudes above 18 km. They serve persistent surveillance, atmospheric research, and low-Earth-orbit (LEO) communication relay roles ^{[1][2]}. Advanced platforms NASA Helios, Airbus Zephyr S, QinetiQ Zephyr S have achieved endurance exceeding 24 hours, with some prototypes demonstrating multi-week sorties using solar arrays, high-energy-density batteries, and fuel cells ^{[3][4]}.

HALE design requires multidisciplinary optimisation. Aerodynamic performance must address low Reynolds number flows where viscous effects dominate ^{[6][7]}, necessitating high-aspect-ratio wings (>20:1) to generate lift while minimising induced drag. Composite materials (carbon-fibre reinforced polymers, glass-fibre composites) achieve required stiffness-to-weight ratios ^{[8][9]}. The energy architecture is hybrid, solar panels feed battery banks for night/low-insolation periods, with PEM fuel cells providing backup power ^{[10][11]}.

Recent advances include AtlantikSolar's 81-hour continuous flight (world record for aircraft <50 kg) ^[39], KARI's EAV-3 (53 kg, 19.5 m wingspan, aspect ratio 17.4) achieving stratospheric flight ^[40], and hybrid turbofan/solar concepts achieving 24-48 hour endurance with 1,000 kg payload ^[41]. Control architectures must be fault-tolerant with redundant actuators and sensors ^{[12][13]}. Aeroelastic coupling in very flexible HALE aircraft requires unified nonlinear models integrating geometrically exact beam elements with the Unsteady Vortex Lattice Method (UVLM) ^[44].

Design gaps persist such as solar cell degradation from radiation and temperature extremes ^{[14][15]}, structural fatigue from repeated load cycles ^{[8][16]}, and photovoltaic integration challenges including micro-cracking, galvanic isolation with carbon-fibre substrates, and manufacturing defects ^[43].

2. Aim and Objectives

2.1 Aim

The aim of this research is to conceive, analyse, and assess a theoretically sound design for a solar-powered HALE UAV. Grounded in aerodynamic, structural, and power-system analyses supported through simulation ^{[1][2][3][4]}, the design integrates state-of-the-art solar photovoltaics, advanced energy storage, and lightweight composites to achieve multi-week endurance with payload versatility and operational reliability.

2.2 Objectives

Primary: Develop a comprehensive conceptual design of a solar-powered HALE UAV satisfying mission endurance, payload, and safety requirements defining airframe geometry, wing planform, propulsion architecture, and power-management strategy, supported through analytical calculations and system-level simulations ^{[5][6][7]}.

Secondary:

Literature Survey - Review existing solar UAVs and HALE systems: aerodynamic configurations, structural materials, energy-storage technologies ^{[1][2][3][4][39][40][41][42][43][44]}.

Design Limitations - Assess energy-storage limitations, low-Reynolds-number aerodynamics, and structural/ fatigue constraints ^{[8][9][10][11][14][15]}.

Performance Evaluation - Employ aerodynamic and power-system simulations to predict endurance, energy budgets, and mission reliability ^{[5][6][7]}.

Design Improvements - Recommend modifications incorporating propulsion optimisation ^[42], aeroelastic analysis ^[44], and photovoltaic integration insights ^[43].

Energy Integration - Quantify renewable energy impact on endurance and reliability, incorporating solar cell challenges ^[43], battery-free concepts ^[45], and charge-margin optimisation ^[39].

3. Market Survey and Literature Review

3.1 Previous Studies

The HALE UAV market has expanded driven by civilian, military, and commercial demand for real-time situational awareness and reduced satellite reliance ^{[1][2][3][4][17]}.

3.1.1 Configuration Studies

Goraj et al. ^[1] presented four aerodynamic concepts for 27 km altitude HALE UAVs, showing carefully designed biplanes achieve comparable efficiency to monoplanes with higher payload capacity. Subsequent work ^[2] addressed low-Reynolds-number aerodynamics, lightweight structures, propulsion, and autonomous flight control. Romeo et al. ^[3] (HELIPLAT programme) demonstrated twin-boom distributed propulsion with solar cell area optimisation. Blended-wing-body (BWB) configurations improve lift-to-drag ratios through distributed load paths ^{[18][19]}. Joined-wing configurations achieve span efficiency factors of 1.4–1.5, reducing induced drag ^[20].

3.1.2 Structural Design

CFRP composites provide necessary stiffness-to-weight ratios for high-aspect-ratio wings ^{[8][9][21]}. Key considerations include weight minimisation, flutter margins, and redundancy ^[22]. Autoclave curing and resin-transfer moulding achieve required material properties ^[21]. Fatigue testing under realistic turbulence conditions is essential for long-duration flight ^[23].

3.1.3 Energy System Analysis

Fuel cells outperform batteries above 2.8 kWh demand ^{[10][11]}, a threshold often exceeded during high-altitude operations (approximately 3.5% power increase per km altitude) ^[10]. Triple-junction GaAs cells achieve 25-30% efficiency; thin-film cells offer lower efficiency but reduced weight ^{[14][15]}. Night-time endurance remains the critical limitation due to battery degradation ^{[24][25]}.

3.1.4 Stability and Control

Longitudinal and directional stability are essential for sustained flight ^{[12][13]}. Flutter analysis is critical for high-aspect-ratio wings as aeroelastic coupling can cause catastrophic failure ^{[22][26]}. Control surfaces require larger deflection angles at low density for equivalent effectiveness ^[27].

3.2 Recent Developments

3.2.1 Perpetual-Flight Solar UAV Design

Oettershagen et al. ^[39] developed the AtlantikSolar UAV (6.9 kg), achieving an 81-hour continuous flight world record for aircraft <50 kg. The conceptual design and analysis framework (CDAF) optimises wing span, aspect ratio, and battery mass using excess time and charge margin metrics. Flight results show

39% minimum state-of-charge, 6.8 hours excess time, and 6.2 hours charge margin. The design philosophy of maximising meteorological robustness is directly applicable to high-altitude platforms.

3.2.2 Solar HALE UAV Aerodynamic Design (EAV-3)

Hwang et al. ^[40] reported KARI's EAV-3: 53 kg (22 kg structure), 19.5 m span, aspect ratio 17.4, cruise speed 6 m/s. Critical considerations included significant wing bending (404 mm at 1-G flight with T-800 carbon fibre) and side-wind effects. No centre-of-gravity shift occurs during flight, yielding a static margin of 28.4% with CG at 31% MAC.

3.2.3 Hybrid Turbofan/Solar HALE UAV Concepts

Mermer and Özgen ^[41] investigated hybrid turbofan/solar HALE UAV design achieving 24+ hours endurance with 1,000 kg payload at 6,096–9,144 m ceiling. Turbofan engines climb to altitude; solar and battery sustain loiter. Competitor studies show average empty weight fractions of 0.6218, aspect ratios of 24.5, and wingspans of 66.45 m.

3.2.4 Propulsion System Optimisation

Dantsker et al. ^[42] presented mission-based propulsion optimisation for the UIUC-TUM Solar Flyer, selecting optimal motor-propeller combinations. The tool achieved 19% efficiency improvement over baseline. This directly affects the power budget and required solar array area.

3.2.5 Aeroelastic Stability of Very Flexible Aircraft

Murua et al. ^[44] developed unified nonlinear aeroelastic-flight mechanics models for very flexible aircraft, integrating geometrically exact beam finite elements with UVLM including free-wake effects. Three-dimensional aerodynamic effects particularly wake interference between wing and tail significantly impact dynamics. This provides flutter boundary prediction more accurately than decoupled approaches.

3.2.6 Photovoltaic Cell Integration on Composite Wings

Łopusiewicz and Książek ^[43] investigated photovoltaic integration challenges:

- **Solar cell performance:** SunPower Maxeon III cells achieve 231–243 Wp/m² with 153 cm² area and 6.5 g weight.
- **Lamination methods:** First method (laminating successive layers to existing fibreglass–photovoltaic composite) shows 9.5% higher power at low loads and 12.5% higher at high loads.
- **Micro-cracking:** Under aerodynamic loading, stress concentrations cause micro-cracking.
- **Galvanic isolation:** Essential with carbon-fibre substrates to prevent short-circuiting.
- **Manufacturing defects:** Electroluminescence testing reveals silicon layer defects, incorrect soldering, lamination imperfections.
- **Efficiency degradation:** Wing deflection reduces photovoltaic output; first lamination method shows superior resistance.

3.2.7 Battery-Free UAV Concepts

Liller et al. ^[45] proposed battery-free solar UAVs storing energy in supercapacitors. Novel energy-aware control algorithms (Greedy and Predictive) prevent brownouts and minimise thrust loss events. While at

a different scale from HALE platforms, the energy-management principles eliminate battery degradation concerns.

3.3 Synthesis and Comparative Analysis

Converging Trends:

1. High-aspect-ratio monoplane configurations dominate (17.4 to 30+ aspect ratio) ^{[40][3]}.
2. Solar cell efficiencies of 25–30% (triple-junction GaAs) are practical upper bounds ^{[14][15][41]}.
3. Hybrid energy storage is necessary for multi-day endurance ^{[10][11][39][41]}.
4. Propulsion optimization yields 15–20% efficiency improvements ^[42].
5. Aeroelastic coupling is critical for high-aspect-ratio flexible wings ^{[22][26][44]}.

Contradictions:

1. **Battery vs. fuel cell:** Original analysis favours fuel cells above 2.8 kWh ^[10], but AtlantikSolar achieved 81 hours with batteries alone ^[39] at lower altitude. Optimal storage depends on platform scale and mission profile.
2. **Configuration philosophy:** EAV-3 aspect ratio of 17.4 ^[40] contrasts with emphasis on >20, suggesting practical constraints limit achievable ratios.
3. **Battery-free approaches:** Liller et al. ^[45] represents a different paradigm but insufficient TRL for HALE.

Configuration	Altitude (m)	Typical AR	Typical span (m)	Energy system	Endurance	Key structural / aero concerns	Primary advantage
<i>Monoplane HALE</i>	12,000–27,000	17–30+	20–70	Solar + battery (for >2.8 kWh)	Multi-day (with hybrid)	High bending, flutter, aeroelastic coupling	Mature, simpler structure
<i>Joined / Box wing</i>	12,000–27,000	effective AR boost (span-eff factor 1.4–1.5)	moderate	Solar + battery	Multi-day (theoretical gains)	Structural complexity, joint loads	Lower induced drag, smaller span for same efficiency
<i>Blended-wing-body (BWB)</i>	cruise alt range	moderate AR	varies	Solar / hybrid	Multi-day possible	Manufacturing complexity, control integration	Improved L/D and payload fraction
<i>Hybrid turbofan + solar</i>	6,000–9,200	~24 (design studies)	~60–70	Turbofan climb + solar/battery loiter	24+ hr (mission-dependent)	Higher empty-weight fraction, integration mass	Payload and climb scalability
<i>Battery-free (supercapacitor)</i>	lower alt (proof-of-concept)	low–mod	small	Supercaps + solar	Short to moderate	Energy buffering, power density limits	Avoids battery degradation (low TRL)
<i>Small perpetual-solar</i>	low–mid alt	very high for scale	10–20 (small UAV)	Solar + batteries	81 hr proven (<50 kg class)	Scale-dependent robustness	Demonstrates meteorological design and control strategies

4. Problem Definition and Design Specifications

The design challenge is synthesising a solar-powered HALE UAV sustaining continuous flight above 18 km for a minimum of 24 hours, with ambition for several days ^{[1][2][3][4]}. The stratospheric environment simultaneously constrains every engineering discipline: aerodynamics must compensate for rarefied air, structures must tolerate large elastic deflections, and the energy system must close its night-time budget without conventional fuel. These constraints are tightly coupled mass added to energy storage increases required wing area, which increases structural mass and power demand in turn.

The conceptual design centres on a high-aspect-ratio monoplane (~32 m span, aspect ratio ~30) with a distributed solar array, a CFRP primary structure, and a hybrid lithium-ion/hydrogen fuel-cell energy architecture. Key performance parameters are summarised in Table 1.

Table 1: Conceptual Performance

Parameter	Value	Basis
Cruise altitude	18-22 km	Mission requirement
Cruise speed	~80 km/h	Aerodynamic sizing
Estimated take-off mass	~114 kg	Mass build-up
Aspect ratio	~30	L/D target 25–30
Cruise power (electrical)	≈ 2 kW	Drag × speed / η_{prop}
Available solar power	≈ 9 kW	Array area × irradiance × η_{cell}
Night storage requirement	≈ 24 kWh	12-hour dark period × 2 kW
Li-ion battery mass (sole storage)	≈ 95 kg	At 250 Wh/kg
Hydrogen mass required	≈ 1.44 kg	PEM fuel cell at 50% efficiency
Total hydrogen system mass	≈ 4.3 kg	Including tank and balance-of-plant
Payload capacity	5-10 kg	Mission specification

The 95 kg battery figure immediately exposes the infeasibility of a battery-only solution on a ~114 kg platform, underpinning the adoption of hydrogen fuel cells as primary long-duration storage.

4.1 Aerodynamic Integration

At 18–22 km, air density falls to roughly one-tenth of sea-level value ($\rho \approx 0.089\text{--}0.119 \text{ kg/m}^3$), forcing low Reynolds number operation in the range $10^5\text{--}10^6$ ^{[6][7]}. Dedicated low-Reynolds-number aerofoils (Eppler E387, Wortmann FX 63-137 series) are required, achieving lift-to-drag ratios of 60-80 at design conditions versus 30-40 for conventional NACA profiles at the same Reynolds number. Induced drag dominates at stratospheric cruise speeds, making high aspect ratio essential; the design targets $AR \approx 30$, consistent with Zephyr-class platforms and the upper bound of the reviewed literature ^{[3][40]}.

Three topologies were evaluated. The conventional high-aspect-ratio monoplane, adopted by all operational HALE platforms (Zephyr S, Helios, EAV-3), is selected for its structural efficiency, straightforward solar cell integration on the upper surface, and established low-Reynolds aerodynamic databases. The biplane explored by Goraj et al. ^[1] achieves comparable efficiency at lower span but suffers performance sensitivity to gap-to-chord ratio and self-shading on solar cells. Blended-wing-body and joined-wing configurations offer L/D improvements and span efficiency factors of 1.4-1.5 ^{[18][19][20]}, but their structural complexity and wake interference effects demand high-fidelity aeroelastic evaluation ^[44] beyond the scope of conceptual sizing.

The wing planform (taper ratio ~ 0.4 , gross area $\sim 32 \text{ m}^2$) gives a wing loading of approximately 35 N/m^2 at 18 km and 80 km/h, consistent with the $30\text{--}50 \text{ N/m}^2$ range observed across comparable solar HALE designs ^{[3][40]}. Aerodynamic screening uses XFOIL analysis across the Reynolds range $3 \times 10^5\text{--}8 \times 10^5$ with CFD validation for three-dimensional effects ^[5].

4.2 Structural Optimisation

Structural design must reconcile ultra-low mass with the bending stiffness needed to carry a $\sim 32 \text{ m}$ span wing through gust and manoeuvre loads. CFRP composites form the primary load path ^{[8][9]}: T-800 intermediate-modulus fibre for spar caps and T-300 woven fabric for wing skins provide a mass-specific stiffness approximately five times greater than aluminium, enabling the slender geometries required ^[8]. The structural mass target is approximately 22 kg for the airframe, consistent with the EAV-3 at 22 kg structure within a 53 kg total ^[40].

Three failure modes govern the design. Static tip deflection is expected to reach 10–15% of semi-span under 1-G flight, introducing geometrically nonlinear load redistribution that must be resolved through large-displacement FEA in ANSYS Workbench ^[5]. Flutter is the critical dynamic threat: very flexible HALE wings exhibit strong bending-torsion coupling that conventional doublet-lattice analysis may overestimate; the Murua et al. ^[44] methodology geometrically exact beam elements coupled with UVLM including wake interference is adopted for higher-fidelity flutter boundary prediction, targeting a flutter margin of $1.2 \times V_D$. Fatigue under multi-day thermal cycling (-50 to -80°C day-night variation at altitude) requires knockdown factors of 0.7 on static allowables and coupon-validated inspection intervals ^{[16][22]}.

4.3 Hybrid Energy System

Solar power is the primary daytime energy source. At 18-20 km, irradiance reaches $\sim 1,360 \text{ W/m}^2$ 36% above sea-level free from tropospheric attenuation ^[15]. Triple-junction GaAs cells at 28% efficiency (within the 25–30% range ^{[14][15]}) are specified for the wing upper surface, with thin-film cells considered for curved fuselage areas. A combined derating of 10-15% is applied for integration losses: micro-cracking under aerodynamic load, wing-deflection cosine penalties, and manufacturing defects identified by electroluminescence testing ^[43]. The first lamination method of Łopusiewicz and Książek ^[43] successive layers onto an existing fibreglass-photovoltaic composite is adopted for its superior resistance to efficiency degradation under both low and high loading. Galvanic isolation between the CFRP substrate and cell interconnects is mandatory to prevent short-circuit paths.

Night-time energy ($\sim 24 \text{ kWh}$) is handled by a two-tier storage architecture. Lithium-ion batteries ($\sim 250 \text{ Wh/kg}$) ^{[10][24]} cover the first 2-4 hours of darkness at higher round-trip efficiency (92–96%); hydrogen PEM fuel cells ($\sim 0.5 \text{ kg/kWh}$ system-level ^{[10][11]}) sustain the remainder, requiring only $\sim 4.3 \text{ kg}$ of total

system mass versus 95 kg of batteries for the equivalent energy. Daytime surplus drives an on-board electrolyser to replenish hydrogen, forming a closed-loop energy cycle. Charge-margin optimization following Oettershagen et al. ^[39] maintaining excess time T_{exc} and charge margin T_{cm} above 6 hours ensures robustness against cloud cover and cell degradation across multi-day operations.

4.4 Mission Profile and Constraints

The UAV climbs from sea level to 18 km over 4–6 hours, timed from dawn to use direct solar power during the energy-intensive ascent ($\sim 4\text{--}6$ kW demand). The Stratospheric loiter is maintained at 18–22 km and ~ 80 km/h in a racetrack pattern, with cruise power of ~ 2 kW. Payload capacity is 5–10 kg accommodating scientific instruments or communications relay hardware with a 50–200 W power budget.

Key environmental constraints include stratospheric winds exceeding 30 m/s during polar vortex conditions ^{[28][29]}, night-time temperatures near -60 °C requiring active battery thermal management, and seasonal variation in solar availability modelled through the mission simulator ^[28]. Cloud cover affects the solar array only during ascent through the troposphere; at cruise altitude the array operates under near-constant stratospheric irradiance. These constraints are incorporated into the mission simulator to define safe operational windows ^{[28][29]}.

5. Methodology

5.1 Literature-Based Foundation

The initial phase reviewed state-of-the-art HALE UAV research ^{[1][2][3][4]}, cataloguing platforms (NASA Helios, Airbus Zephyr, DARPA Vulture, KARI EAV-3 ^[40]) and mission parameters. Solar-cell literature was examined (triple-junction vs thin-film ^{[14][15]}); energy-storage options were assessed (lithium-ion vs fuel cells ^{[10][11][24][25]}).

Expanded literature includes perpetual-flight methodologies ^[39], propulsion optimisation ^[42], coupled aeroelastic modelling ^[44], photovoltaic integration challenges ^[43], hybrid turbofan/solar concepts ^[41], and battery-free architectures ^[45].

5.2 Mission Requirements Definition

Mission requirements: minimum 24-hour endurance, target L/D 25–30, payload for scientific instruments and communications, structural stiffness for flutter avoidance ^{[1][2][3][4]}. EAV-3 ^[40] validates target L/D achievable; AtlantikSolar ^[39] confirms energy balance closure at low altitude, extensible to high altitude ($\sim 1,360$ W/m² irradiance).

5.3 Aerodynamic Modelling

5.3.1 Airfoil and Wing Analysis

Aerodynamic evaluation couples analytical theory with computational fluid dynamics (CFD) ^{[5][6][7]}. Candidate airfoils screened using low-Reynolds-number data (C_L , C_D , C_L/C_D across $\sim 0.5\text{--}2 \times 10^6$ Reynolds range). CFD captures 3D effects: tip vortices, spanwise flow, pressure-gradient separation. EAV-3 ^[40] demonstrates wing bending and side-wind consideration at low cruise speeds (6 m/s).

5.3.2 Wing Configuration Comparison

Three configurations were examined—biplane, BWB, joined-wing^{[1][2][3][18][19][20]}. The biplane achieves a lower effective aspect ratio through interference. BWB provides smooth pressure distribution and higher L/D. Joined-wing yields 1.4–1.5 span efficiency^[20]. Mermer and Özgen^[41] competitor study shows average aspect ratios of 24.5 and wingspans of 66.45 m; their structural weight prediction model warrants adoption.

5.4 Structural Modelling

5.4.1 Material Selection

CFRP selected as baseline^{[8][9][21]}. Anisotropic nature allows stiffness and strength tailoring through fibre ply orientation along principal load paths. Out-of-plane reinforcement and nano-engineered resin systems improve damage tolerance^[8].

5.4.2 Structural Analysis

Detailed finite-element model constructed using ANSYS Workbench^[5], incorporating aerodynamic pressure loads from CFD and inertial loads from flight dynamics. Flutter analysis uses a coupled aeroelastic module integrating aerodynamic influence coefficients (doublet-lattice method) with structural modal data. Murua et al.^[44] framework offers higher-fidelity flutter prediction for very flexible wings where geometrically nonlinear deformations alter aerodynamic characteristics.

5.5 Energy System Modelling

5.5.1 Solar Power Estimation

Solar subsystem sized based on triple-junction photovoltaic cells exceeding 350 W/kg specific power under stratospheric irradiance ($\sim 1,360 \text{ W/m}^2$)^{[14][15]}. The model incorporates cosine losses, shading, and seasonal/diurnal variations at 18–20 km.

Łopusiewicz^[43] integration challenges factored in: first lamination method shows 9.5% higher power under low loading; efficiency degradation under wing deflection measurable. These losses and manufacturing defect allowances become derating factors.

5.5.2 Battery/Fuel-Cell Simulation

Hybrid storage model simulates lithium-ion pack ($\sim 250 \text{ Wh/kg}$) and PEM fuel-cell stack in MATLAB/Simulink^{[10][11]}. Night-time endurance is evaluated by integrating net power deficit over dark periods.

AtlantikSolar charge-margin methodology^[39] provides a safety margin framework. Excess time (T_{exc}) and charge margin (T_{cm}) metrics quantify margin above minimum acceptable threshold. Designing for a 30–40% charge margin (as demonstrated in an 81-hour flight) ensures robustness against cloud cover, degradation, and unexpected demands.

5.5.3 Hybrid Energy Management

The energy-balance model orchestrates power flows from solar array to storage and propulsion/avionics^{[10][11]}. Hierarchical management prioritises battery charging with excess solar, fuel-cell supplementation during extended night.

5.6 Stability and Control Analysis

Stability assessment builds on classical longitudinal and directional theory, adapted for low-density HALE flight ^{[12][13]}. Static margin calculated by locating the aerodynamic centre and comparing to CG location, ensuring minimum 5% MAC margin. EAV-3 ^[40] provides reference: static margin 28.4% with CG at 31% MAC.

Control surface sizing uses a derivative method, incorporating reduced dynamic pressure effects ^[27]. Flutter boundaries estimated using Doublet Lattice Method (DLM), with UVLM option ^[44] for detailed assessment.

5.7 Performance Simulation

Integrated simulations verify conceptual UAV meets all targets. Altitude–power relationship incorporates 3.5% power increase per km altitude gain. MATLAB used for system-level analyses; ANSYS for high-fidelity aero-structural coupling ^[5].

5.8 Design Integration and Optimisation

Multi-objective genetic algorithm (MOGA) explores design space, simultaneously minimising mass, maximising endurance and L/D while satisfying flutter margin, structural stress, and solar-panel constraints. Optimiser iteratively adjusts wing span, chord distribution, spar geometry, CFRP lay-up, solar-cell layout, and storage sizing.

6. Analysis and Discussion

6.1 Aerodynamic Performance Assessment

High-aspect-ratio wings are essential for target L/D of 25-30. Literature shows consistent aspect ratios from 17.4 (EAV-3) to 30+ (Zephyr-class), with lower end more practical for manufacturing and ground handling ^[40].

Joined-wing configurations offer 1.4-1.5 span efficiency ^[20], significant advantage over conventional designs. However, structural complexity and wake interference require high-fidelity aeroelastic evaluation ^[44].

Low cruise speeds (6-8 m/s for EAV-3) for optimal solar power present challenges: atmospheric disturbance sensitivity, larger control surfaces for authority, significant wing bending accommodated structurally ^[40].

6.2 Energy System Feasibility

Numerical validation demonstrates hybrid energy storage criticality. Night-time energy requirement (~23.9 kWh for 12-hour dark period) presents fundamental challenge:

- **Battery Solution:** At 250 Wh/kg, required mass ~95.7 kg exceeding feasible limits for ~114 kg empty weight platform.
- **Hydrogen Fuel Cell:** At 33.3 kWh/kg (LHV) and 50% system efficiency, only 1.44 kg hydrogen required. Including tankage and balance-of-plant (3× multiplier), total ~4.3 kg dramatic reduction versus batteries.

6.3 Propulsion Optimisation Potential

Dantsker et al. ^[42] demonstrates 19% efficiency improvement from motor-propeller matching. Applying the present design reduces electrical power from ~1,994 W to ~1,615 W corresponding night-time storage reduction from 23.9 kWh to ~19.4 kWh.

Mass savings: battery reduces from 95.7 kg to ~77.6 kg; hydrogen from 1.44 kg to ~1.16 kg (~3.5 kg system mass).

6.4 Structural and Aeroelastic Considerations

CFRP composites are essential for stiffness-to-weight ratios. EAV-3 demonstrates 404 mm wing bending at 1-G flight achievable with T-800 carbon fibre ^[40].

High-aspect-ratio very flexible wings introduce significant aeroelastic coupling. Conventional decoupled approaches may underpredict flutter boundaries. UVLM-based methodology ^[44] provides an accurate framework for wing-tail wake interference effects.

6.5 Photovoltaic Integration Challenges

Łopusiewicz and Książek ^[43] practical challenges must be factored:

1. **Micro-cracking:** Aerodynamic loading causes stress concentrations leading to efficiency reduction.
2. **Galvanic isolation:** Carbon-fibre substrates require proper isolation to prevent short-circuiting.
3. **Manufacturing defects:** Electroluminescence reveals 5–10% cell defects requiring quality control.
4. **Efficiency degradation:** Wing bending reduces photovoltaic output.

Conservative 10-15% derating factor recommended for theoretical solar power calculations.

6.6 Comparison with Established Platforms

Table 1 presents the conceptual design against established HALE platforms.

Platform	Mass (kg)	Wingspan (m)	Aspect Ratio	Endurance
Zephyr S	53	25	24	25+ days
Helios	730	75	30	14+ hours
EAV-3	53	19.5	17.4	Stratospheric test
AtlantikSolar	6.9			81 hours
Present Design	~114 (airframe)	32	~30	Target: 24+ hours

Design targets align with literature benchmarks; hybrid energy storage provides a practical multi-day endurance pathway.

7. Conclusion and Future Work

The study indicates technical feasibility and multidisciplinary complexity in conceptual design of a HALE UAV powered by renewable energy. Expanded literature review establishes HALE reliance on aerodynamics, lightweight structures, and high-efficiency energy systems for sustained flight above 18 km^{[1][2][3][4][39][40][41][42][43][44]}.

Numerical evaluation suggests realistic power demands (~2 kW electrical) with high-altitude solar flux and triple-junction GaAs providing ~9 kW usable power. Night-time storage (~23.9 kWh) makes pure lithium-ion batteries infeasible (~95 kg); hydrogen fuel-cell storage meets requirements at <5 kg. Propulsion optimisation^[42] could reduce power demands by up to 19%.

Expanded literature strengthens design foundation:

1. **Perpetual-flight methodology**^[39]: Rigorous energy-system sizing and charge-margin framework.
2. **EAV-3 empirical validation**^[40]: Confirms high-aspect-ratio flexible wing feasibility.
3. **Hybrid propulsion concepts**^[41]: Viable pathways for payload-intensive HALE.
4. **Advanced aeroelastic analysis**^[44]: Accurate flutter prediction for flexible wings.
5. **Photovoltaic integration research**^[43]: Quantifies efficiency losses, establishes manufacturing practices.
6. **Battery-free concepts**^[45]: Future energy paradigms eliminating degradation.

Future Work:

- Detailed computational models incorporating atmospheric variations (turbulence, temperature gradients, cloud shading).
- Prototype-scale CFD and wind-tunnel validation under low-Reynolds conditions.
- Composite wing segment testing for flutter boundaries and material reliability.
- Energy management algorithm development incorporating charge-margin methodology^[39].
- Propulsion optimization using Dantsker framework^[42].
- Coupled aeroelastic-flight dynamic application^[44] for flutter analysis.
- Photovoltaic integration best practices adoption^[43] for wing manufacturing.
- Scaled demonstrator fabrication for flight testing validation.
- Emerging technology research: morphing wings, solid-state batteries, ultra-light flexible solar arrays, supercapacitor storage^[45].

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New Citations (Added in Version 2)

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